AI Exam Answer Grader

A Reusable NLP Asset for Rubric-Based Descriptive Answer Assessment

Final Project Report | SNLP Course Project 2026 — AI University: Building Reusable NLP Assets
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Code: github.com/RitwikShetty05/ai-exam-answer-grader | 31 October 2026

Abstract. We built an advisory grader that recommends a mark, explainable feedback and a confidence score for a student's descriptive answer, given the question, the teacher's reference answer and an optional rubric. It combines a fine-tuned cross-encoder that measures how close the answer's meaning is to the reference with a local LLM that checks every rubric point and must quote the student's own words as evidence; the two are calibrated to the teachers' marking scale and averaged, and their agreement gives the confidence. Uncertain or suspicious answers are routed to a teacher. On 450 Mohler answers to questions the models had never seen, the final grader is 0.57 marks out of 5 from the teachers on average (TF-IDF baseline 0.72, mean predictor 0.89; two human teachers differ by 0.72), significantly better than the best similarity model alone. The grader is packaged as a Python module and a FastAPI service, accepts the output of Group 1's Handwritten Exam Digitizer directly, and has a Django + React teacher application that records every teacher decision. On Group 1's handwritten pages, neat handwriting barely changes our marks, while cursive handwriting does and is flagged for a teacher.

1. Problem Statement

Marking descriptive exam answers is slow, and two teachers often give the same answer different marks: in our dataset the two graders differ by 0.75 marks out of 5 on average and by 2 marks or more on 18% of answers. We want a component that gives teachers a fast, consistent and explainable first-pass mark for each answer, says how sure it is, and leaves the final decision to the teacher. The handout forbids simply asking an LLM to "give marks", so the component must contain a genuine NLP pipeline and be evaluated against human marks.

2. Institutional Context

Users are faculty and teaching assistants (first-pass marking of answer scripts), the examination cell (consistency checks between evaluators and re-evaluation requests), and other AI University components. In the AI University architecture the grader belongs to Academic Services → Assessment. Its main partner is Group 1's Handwritten Exam Digitizer, which turns scanned answer sheets into text that our grader marks; other services (for example Group 12's malicious-input detector) can sit in front of it through the same API.

3. Existing Approaches

Automatic short-answer grading (ASAG) has moved through several eras (Burrows et al., 2015). Early systems matched keywords and concept patterns; Mohler et al. (2011) combined lexical and knowledge-based semantic similarity with dependency-graph alignment and released the dataset we use. Sentence embeddings such as Sentence-BERT (Reimers and Gurevych, 2019) compare meaning rather than words, and fine-tuned transformer cross-encoders that read the reference and the student answer together gave the next large gains (Sung et al., 2019). Most recently, large language models have been used to grade with rubrics, but their marks can be inconsistent, strict or lenient, and open to prompt injection, so they need calibration and human oversight. Commercial tools mostly assist rather than replace the teacher: ETS's e-rater scores essays alongside human raters (Attali and Burstein, 2006), and Gradescope groups similar answers so a teacher can mark them together. Our approach follows the assistive model and combines a fine-tuned cross-encoder with an evidence-checked LLM.

4. NLP Formulation

We formulate grading as supervised regression on a text pair: given (reference answer, student answer) and optionally rubric concepts, predict the teachers' mark on a 0–5 scale, then scale it to the question's maximum marks. A second, structured task is concept classification: for each rubric concept, decide present, partially present or missing, with the supporting span from the student's answer (an extractive rationale). The confidence is a calibrated probability that the predicted mark is within ±1 mark of a teacher's mark. Success is measured with mean absolute error (MAE), RMSE, agreement within ±1 mark and Pearson correlation, always compared with a mean-predictor floor.

5. Dataset

Mohler short-answer grading dataset (Kaggle mirror abdokamr/mohler): answers to data-structures and C++ questions from a university computer science course, each marked 0–5 by two teachers; our label is their average. After removing 9 answers stored as "#NAME?" (a spreadsheet error that deleted the text), 2,264 answers to 79 questions remain. Answers are short (19 words on average; 95% have 45 words or fewer) and marks are skewed high (mean 4.17), so always predicting the average is already within ±1 mark about 80% of the time. Full documentation is in Appendix B.

Group 1's data (Week 7): two real photographed answer sheets on computer networks with hand-typed true text, from Group 1's repository, read with their digitizer. They have no teacher marks, so they are used to study integration and OCR effects, not accuracy.

6. Preprocessing

- **Cleaning:** corrupted answers removed; text normalised (lower-casing, punctuation and stop-word handling, lemmatisation) for the TF-IDF baseline only; neural models see the raw text.
- **Answer checks before grading:** blank answers ("not answered", "I don't know") get 0 without calling the models; instruction-like text is flagged; tags typed by the student are neutralised before the answer is placed in the LLM prompt.
- **Rubrics:** the teacher's rubric if given, otherwise 1–5 key concepts generated once per question from the reference answer by the LLM and saved for review.
- **Group 1 pages (adapter):** the name and roll-number header is removed; the question number is found even through OCR mistakes ("QL", "@1]" → Q1); pages are split by question and joined when an answer continues; the lowest OCR confidence is kept; a missing student ID adds a review note.

7. Baseline

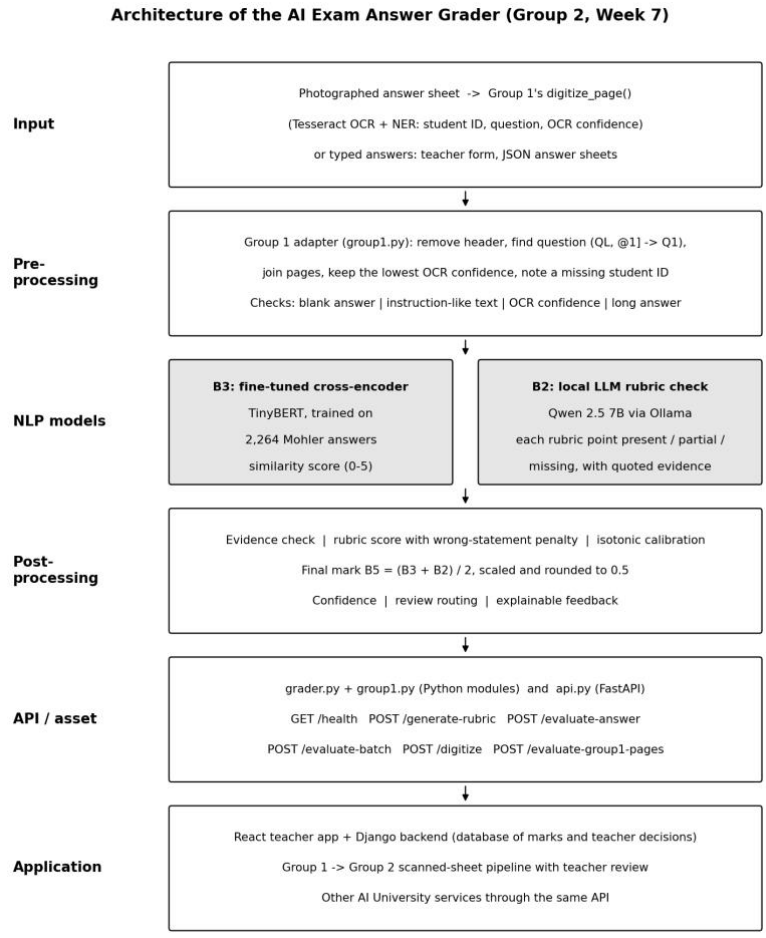
Approach A (Weeks 1–2) represents the reference and student answers as TF-IDF vectors, takes their cosine similarity plus a keyword concept-coverage score, and maps it to marks with a fixed linear formula. It was off by 2.49 marks on average because the fixed scale put almost every mark too low; learning the mapping from the human marks (calibrated TF-IDF) cut the error to 0.72. Its key weakness is lexical: a correct answer in different words ("log n" for "the height of the tree") shares no words with the reference.

8. Proposed Method

- **B1 — sentence embeddings:** MiniLM and MPNet bi-encoders compare meaning vectors, with a learned mapping to marks.

- **B3 — fine-tuned cross-encoder:** TinyBERT (cross-encoder/stsb-TinyBERT-L4; Jiao et al., 2020) reads the reference and student answer together and was fine-tuned on (reference, answer) pairs to predict mark ÷ 5 (3 epochs, batch 16, learning rate 4e-5, binary cross-entropy).
- **B2 — evidence-checked LLM rubric grader:** Qwen 2.5 7B Instruct (Qwen Team, 2024), run locally with Ollama (temperature 0, fixed seed, JSON-schema output). For each rubric concept it gives present, partial or missing with a quote from the student's answer; our code accepts the quote only if at least 80% of its words occur in the answer, matches the checks to the rubric, and computes the rubric score itself (present 1, partial 0.5, minus half a point per wrong statement). The LLM is much stricter than the teachers, so the average of its score and the rubric score is mapped to the teachers' scale with isotonic regression (Zadrozny and Elkan, 2002).
- **B5 — final grader:** the average of B3 and calibrated B2, scaled to the maximum marks and rounded to 0.5.
- **Confidence:** a logistic regression on the disagreement between B3 and B2 and between the LLM's own score and the rubric score, estimating the chance of being within ±1 mark of a teacher.
- **Review routing:** an answer goes to a teacher if confidence < 0.60, if the LLM and rubric scores differ by 2 or more, if instruction-like text is found, if OCR confidence < 0.80, or if the answer is longer than 100 words (outside the similarity model's training range).

9. Architecture



System architecture: input → pre-processing → NLP models → post-processing → API/asset → applications.

10. Experiments

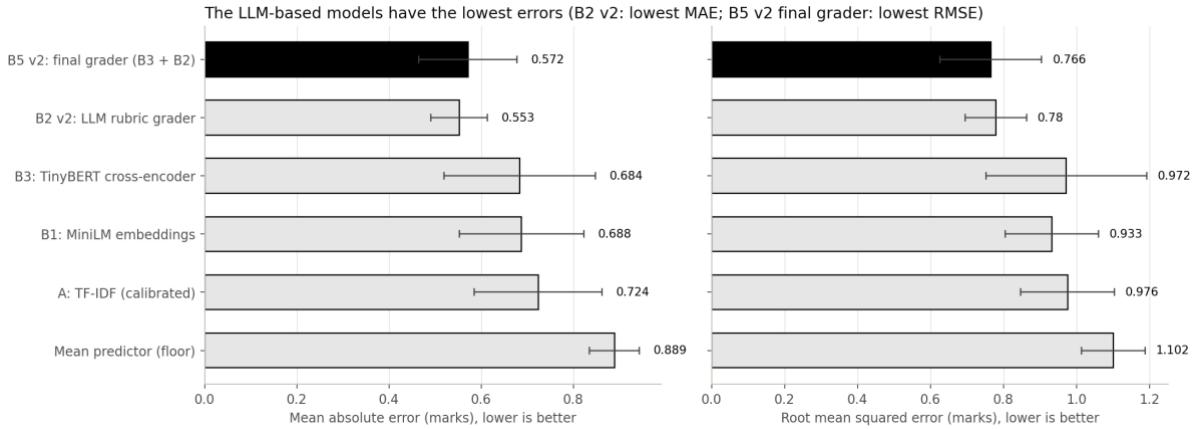
- **Protocol:** 5-fold cross-validation grouped by question (seed 42), so every prediction is on questions the model never saw — the real use case of a new exam question. Results are mean ± SD over folds.
- **LLM evaluation sample:** the LLM needs about 8 seconds per answer, so B2 and B5 were evaluated on a stratified sample of 450 answers (all 309 answers marked below 3, plus 141 others) with sampling weights; all methods were re-scored on the same 450 answers. Ten answers used while building the prompt were excluded.
- **Significance:** paired bootstrap over questions (2,000 resamples); a difference is called real only if the 95% interval excludes 0.
- **Robustness:** spelling mistakes, irrelevant padding, repetition and truncation of 10 strong answers; fake answers (copied reference, copied question, keyword lists, "I don't know") for 11 questions; five prompt-injection attacks on 10 weak answers.
- **Group 1 integration (Week 7):** Group 1's digitizer on their two real pages; and OCR-style errors added to the 95 fold-0 answers at Group 1's measured character error rates (0.04 neat, 0.19 average, 0.34 cursive), graded with a similarity model retrained without fold 0 (570 gradings).

11. Results

Method (450-answer sample, weighted)	MAE ↓	RMSE ↓	Within ±1 ↑	Pearson r ↑
Floor: mean predictor	0.889 ± 0.054	1.102 ± 0.087	79.7%	–
A: TF-IDF (calibrated)	0.724 ± 0.139	0.976 ± 0.128	69.6%	0.463
B3: fine-tuned cross-encoder	0.684 ± 0.164	0.972 ± 0.220	74.6%	0.527
B4: ensemble B3 + MiniLM	0.665 ± 0.107	0.902 ± 0.139	76.0%	0.574
B2: LLM score (raw)	1.367 ± 0.223	1.787 ± 0.162	49.3%	0.659
B2 v2: LLM rubric grader (calibrated)	0.553 ± 0.061	0.780 ± 0.084	83.5%	0.709
B5 v2: final grader (B3 + B2 v2)	**0.572 ± 0.107**	**0.766 ± 0.139**	**81.9%**	**0.714**

On all 2,264 answers (Weeks 3–4): mean predictor 0.909, TF-IDF calibrated 0.724, MiniLM 0.700, MPNet 0.701, B3 0.634 ± 0.108. The sample estimates carry about ±0.05 marks of sampling noise.

- **Real but modest gains.** B3 beats calibrated TF-IDF on all answers (0.091 marks, 95% CI 0.034–0.146). The final grader beats B3 (0.110, CI 0.048–0.175). The Week 6 fixes added only 0.015 marks over Week 5 (not significant). B2 v2 has the lowest MAE and B5 v2 the lowest RMSE and highest correlation; their difference is small.
- **Against humans.** The two teachers differ by 0.72 marks on the sample; the final grader is 0.60 from teacher 2 and 0.82 from the stricter teacher 1. It was trained on their average, which favours it, so this does not show it is as good as a teacher.
- **Where it struggles.** On weak answers (marked below 2.5) the error is 1.42 marks (B3 1.50; the raw LLM, which judges correctness, 1.02); on strong answers 0.43.
- **Confidence and routing.** Answers given confidence ≥ 0.85 (59% of answers) were within ±1 mark 87% of the time; below 0.50 (4.5%), only 27% (AUC 0.62). The review rules send 18.8% of answers to a teacher and catch 45.7% of the large errors (more than 1.5 marks).
- **Robustness.** The LLM takes marks off for spelling (−0.85) and padding (−1.20); keyword lists fool B3 (4.81 / 5) but much less the LLM (2.91). In Week 5, 4 of 50 injected answers got full marks from the raw LLM; the final pattern check flags 40 of 50 with no false alarms on the 2,264 real answers.



MAE and RMSE of the main methods (error bars: SD over 5 folds); the final grader is shown in black.

Week 7: OCR errors (95 answers)	CER	MAE vs teachers	Within ±1	Mark change	Flagged
Clean text	0.00	0.567	75.6%	–	23.1%
Group 1 neat-page error rate	0.04	0.659	76.0%	0.22	17.8%
Group 1 average error rate	0.19	1.154	47.2%	0.85	39.0%
Group 1 cursive-page error rate	0.34	1.344	33.9%	1.17	37.9%

The similarity model's score moved by 0.20 / 1.13 / 1.67 marks and the LLM's by 0.28 / 0.67 / 0.78, so character errors hurt the similarity model most. An OCR-aware LLM prompt gave no significant gain (+0.024 and −0.088 marks), so it is off by default. On Group 1's real pages, the neat page (CER 0.04, OCR confidence 0.82) got the same mark from the OCR text as from the true text; the cursive page (CER 0.35, confidence 0.40) changed by 1 mark and was flagged for four reasons.

12. Error Analysis

Twelve errors were analysed in Week 5 and grouped into six causes; ten are shown, plus the main Week 7 failure. Marks are out of 5.

Question / answer	Marks	Cause	Why the system failed
Queue: "like a linked list expect first in last out"	Human 1.25; B3 4.70; final 3.0	Wrong on the right topic	Same words as the reference, but describes a stack. The LLM gives 0, yet calibration lifts 0 to 2.8 and B3 agrees, so confidence stays high (our clearest remaining failure).
Array dimensions: "just the first one at least"	Human 2.0; B3 4.86	Wrong on the right topic	The reference says all dimensions except the first; same words, opposite meaning.
BST search: "log n"	Human 4.75; B3 1.13	Correct in other words	No word overlap with "the height of the tree"; the LLM only gave "partial". Final mark now 4.5.
Testing stages: "coding and refining"	Human 5.0; B3 3.14	Correct in other words	A correct three-word answer to a two-sentence reference.
Merge sort, correct answer	Human 5.0; B2 raw 2.0	LLM too literal	Every concept "implied, not explicit"; a spelling mistake listed as a wrong statement.
Queue with an array	Human 4.0; B2 raw 0.0	LLM too literal	Rubric split into literal parts; a correct sentence called wrong.
Inorder traversal, order reversed	Human 1.5; B2 raw 4.5	LLM inconsistent	All concepts marked present although its own feedback says the order is reversed.
BST search: "depends on the location of the node"	Human 1.9; B2 raw 4.5	LLM changed rubric	The LLM rewrote the concept to match the answer (fixed in Week 6 by matching checks to the original rubric).
"not answered"	Human 0; B2 calibrated 2.89	Calibration floor	Linear calibration never went below 2.8 (fixed by the blank-answer rule).
C++ start: "At the root"	Grader 1: 0; grader 2: 5	Labels disagree	Wrong answer; the label is the average 2.5, so a correct 0 looks like a large error.
Group 1 page on the OSI model (Week 7)	Ours 8 / 10; covers ≤ 3 of 7 points	New course, long answer	Mohler calibration is too lenient for a new course and B3 reads only half of a long page; flagged as a long answer.

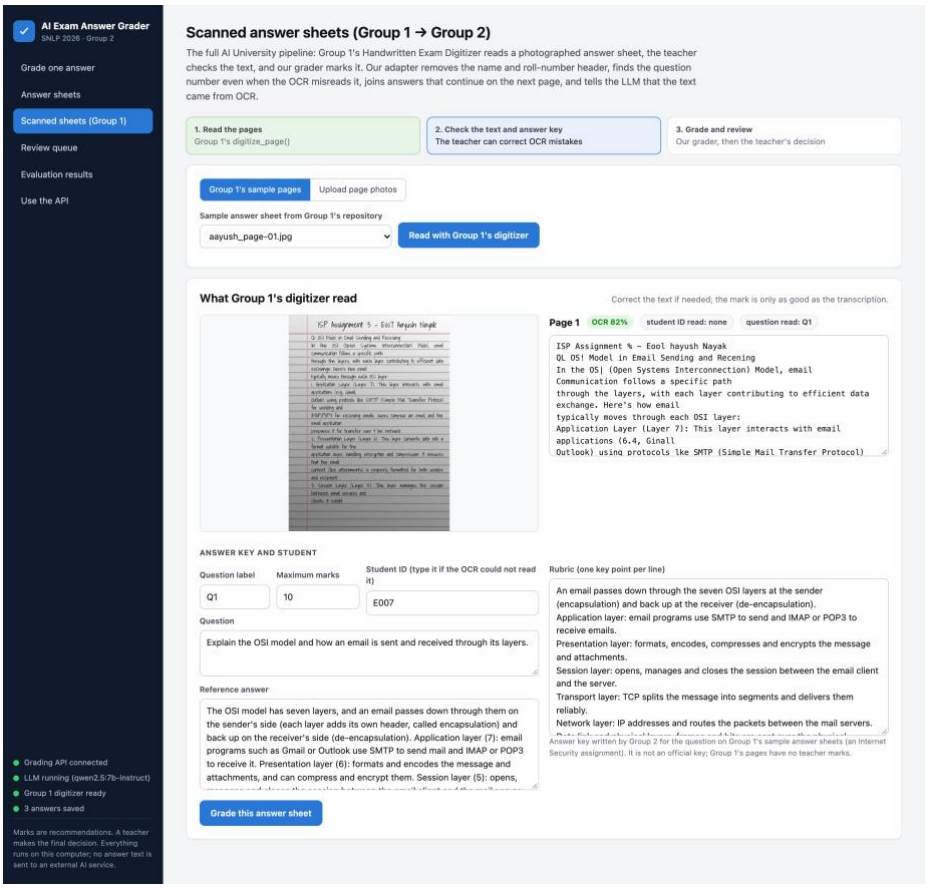
13. Reusability

The asset is a Python module (`src/grader.py`, `src/group1.py`) and a FastAPI web service (`src/api.py`) with interactive documentation at /docs. Any program can call it with JSON over HTTP:

Endpoint	Purpose
`GET /health`	Service, LLM and Group 1 digitizer status
`POST /generate-rubric`	Rubric concepts from a reference answer, for a teacher to check
`POST /evaluate-answer`	One answer → score, feedback, confidence, review flags, concept checks
`POST /evaluate-batch`	Answers in the handout's Group 1 format (student_id, question, answer_text, page, confidence)
`POST /digitize`	One scanned page → Group 1's digitize_page() record
`POST /evaluate-group1-pages`	Group 1's page records exactly as returned → graded answers

The Django + React teacher application is itself a client of this API: it grades single answers, answer sheets and scanned pages (through Group 1's digitizer, with an editable transcription), keeps a review queue,

stores every recommendation and teacher decision in a database, and exports marks as CSV. Full input/output documentation is in Appendix A; the repository README explains installation and use.



The scanned-sheet page: Group 1's digitizer output, its OCR confidence and the answer key, before grading.

14. Responsible AI

Privacy (local models; no answer text leaves the computer), bias (spelling and handwriting), hallucination (evidence-checked rubric points), explainability (concept-by-concept feedback with the student's words), security (prompt-injection defences) and human oversight (advisory marks, review queue, stored teacher decisions) are covered in the one-page statement in Appendix C.

15. Limitations

- Evaluated on one public dataset (one course, lenient teachers, short answers) and a 450-answer sample; the Week 6 fixes and review thresholds were designed on the same data.
- The calibration floor (about 2.8) lets wrong answers on the right topic score about 3 out of 5, and the calibration does not transfer to a new course without teacher-marked examples.
- The confidence score separates good and bad marks only moderately (AUC 0.62); the review rules catch 46% of large errors.
- The 7B LLM lacks some subject knowledge, penalises spelling and takes 6–8 seconds per answer; all rubric concepts weigh the same; the similarity model reads at most 256 word-pieces.
- Group 1 integration was tested on two real pages and simulated OCR errors; handwritten student IDs are rarely read.

16. Future Work

1. Collect anonymised answers and teacher marks from our own university to test and recalibrate the grader per course.
2. Weighted rubric points and a cap on the mark when the LLM finds a wrong statement about the main idea (the queue failure).
3. Handle long answers by grading them in parts, and use Group 1's per-question records when they add question splitting.
4. Production deployment: teacher logins and roles, PostgreSQL instead of SQLite, a job queue for large batches, and a larger LLM on a college GPU server — all still on premises for privacy.
5. Use teachers' corrections from the review queue to retrain the models over time, with regular bias checks.

References

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Sung, C., Dhamecha, T. and Mukhi, N. (2019). Improving short answer grading using transformer-based pre-training. Proceedings of AIED 2019.

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APPENDICES

Appendix A. API Documentation

Start the service with `cd src` and `uvicorn api:app --port 8000` (Ollama must be running for the LLM part). Interactive documentation with every field: `http://localhost:8000/docs`.

POST /evaluate-answer — input

Field	Type	Required	Meaning
question_text	text	yes	The exam question.
reference_answer	text	yes	The teacher's model answer.
student_answer	text	yes	The student's answer (answer_text from Group 1).
maximum_marks	number	no (5)	The score is scaled to this and rounded to 0.5.
rubric	list of text	no	Key concepts, all worth the same; generated if missing.
question_id, student_id	text	no	Labels returned unchanged.
ocr_confidence	0 to 1	no	From Group 1; below 0.80 the answer is flagged.

Output

Field	Meaning
score, maximum_score	Recommended (advisory) mark, scaled to the maximum marks.
feedback	Covered, partly covered and missing concepts, the first wrong statement, and a short comment.
confidence	Chance (0–1) that the score is within 20% of the maximum marks of a teacher's mark.
needs_human_review, review_reasons	Whether a teacher should check the answer, and why.
concepts	Each rubric concept with status (present, partial, missing) and the student's words as evidence.
wrong_statements	Statements the LLM judged to be factually wrong.
rubric_source, grading_mode	teacher / saved / generated; full / similarity_only (LLM down) / blank_answer.
scores_out_of_5	Component scores: similarity_model, llm, rubric_before_penalty, wrong_statement_penalty, rubric, llm_rubric_average, llm_calibrated, final.
pages, answer_text, removed_header	Group 1 page requests only: pages used, graded text, removed header.

Example

```
import requests

request_body = {
    "question_text": "How many steps does it take to search a node in a binary search tree?",
    "reference_answer": "The height of the tree.",
    "student_answer": "log n",
    "maximum_marks": 5,
}
reply = requests.post("http://localhost:8000/evaluate-answer", json=request_body)
print(reply.json())

# {"score": 4.5, "maximum_score": 5, "confidence": 0.9, "needs_human_review": false,
#  "feedback": "Only partly covered: It depends on the height of the tree. | ...",
#  "concepts": [{"concept": "It depends on the height of the tree.", "status": "partial", "evidence": "log
n"}], ...}
```

Errors: a missing field returns HTTP 422 naming the field; an empty reference answer or an unknown question label returns 400; an LLM failure after retries returns 503; if the LLM is not running, the similarity-model mark is returned with confidence 0.75 and a review flag.

Appendix B. Dataset Documentation

Item	Details
Source	Mohler, Bunescu and Mihalcea (2011) short-answer grading dataset; Kaggle mirror abdokamr/mohler (public).
Content	Answers by undergraduate students to 79 questions (data structures and C++), each with a reference answer; two teachers marked every answer 0–5.
Size	2,273 answers; 9 removed ("#NAME?" spreadsheet error); 2,264 used. Label = average of the two marks (mean 4.17).
Preprocessing	Corrupt rows removed; TF-IDF baseline uses lower-casing, stop-word handling and lemmatisation; neural models use raw text; blank answers detected by rule.
Splits	5 folds grouped by question (15–16 questions per fold), seed 42; LLM evaluated on a stratified 450-answer sample with sampling weights.
Limitations	One course, English only, lenient marking, short answers (95% ≤ 45 words), 18% of answers with a 2-mark disagreement between the teachers; no handwriting.
Group 1 pages	Two photographed answer sheets on the OSI model with hand-typed true text (Group 1's repository); no teacher marks; our own 7-point answer key.

Appendix C. Responsible AI Statement

- **Privacy:** the evaluated system runs entirely on the teacher's computer: Group 1's OCR, our cross-encoder and the LLM (through Ollama). No answer text is sent to an external AI service. Uploaded page photos are deleted after reading; the database stores only answer text, marks and teacher decisions, and must stay on college systems. The student's name header is removed before grading. The optional online demo uses a hosted LLM and is limited to the public sample answers.
- **Bias:** the LLM takes marks off for spelling mistakes and irrelevant text, and OCR errors depend on handwriting (the cursive page had nine times the error rate of the neat one), which could disadvantage second-language writers and students with untidy handwriting. The similarity model favours wording close to the reference. Mitigations: an explicit instruction not to penalise spelling, low OCR confidence always routes to a teacher, the teacher can correct the transcription, and we report results by answer quality.
- **Hallucination:** the LLM must quote the student's words for every concept it credits, and the code rejects quotes that are not in the answer; the rubric score is computed by code, not the LLM; a large gap between the LLM's score and the rubric score sends the answer to a teacher. Remaining risks: inconsistent reasoning, correct sentences or OCR garbage listed as wrong statements.
- **Explainability:** every mark comes with the rubric points covered, partly covered and missing, the student's own words as evidence, wrong statements, component scores and a chart of how the mark was calculated.
- **Security:** the answer is isolated between tags, the prompt says before and after it that its content is never an instruction, typed tags are neutralised, and nine injection patterns are flagged (40 of 50 attack answers, no false alarms on real answers). File uploads are restricted to images and Django's CSRF protection is on. New kinds of attack may still get through.
- **Human oversight:** marks are recommendations only. Answers with low confidence, LLM–rubric disagreement, instruction-like text, low OCR confidence or unusual length wait in the review queue; the teacher's mark and comment are stored next to the recommendation and are the final mark.
- **Limitations of use:** not validated for other courses, languages or long essays; not to be used for high-stakes marking without a teacher reviewing every flagged answer and a sample of the others.

Appendix D. Team Contributions and Timeline

Member	Part	Main work
E008 Ritwik Shetty	Problem, data and baseline	Proposal and use case, Mohler dataset and cleaning, TF-IDF baseline, metrics, mean-predictor floor
E010 Dhruvit Thakkar	Improved similarity models	Sentence embeddings, cross-encoder and fine-tuning, calibration, grouped cross-validation, ensemble, significance tests
E011 Ajinkya Hatle	LLM rubric grader	Local LLM, rubric generation, evidence check, rubric score and penalty, blank rule, isotonic calibration, final ensemble
E012 Bharat Baswan	Evaluation, error analysis and safety	Stratified sample and weights, human agreement, error analysis, robustness and prompt injection, confidence, review rules, responsible AI
E009 Chaitanya Thakur	API, interface and integration	Python module, FastAPI service, Django + React application, Group 1 adapter and endpoints

Weeks	Deliverable
1–2	Proposal, Mohler data, TF-IDF baseline and first metrics
3–4	Sentence embeddings, cross-encoder, fine-tuning, grouped 5-fold CV, ensemble, bootstrap tests
5	Local LLM rubric grader, 450-answer evaluation, 12 explained errors, robustness and prompt-injection tests
6	Fixes from the error analysis, confidence score, review routing, module, FastAPI service, documentation, demo
7	Group 1 integration (adapter, endpoints, OCR experiment), Django + React application, responsible AI
8	Final testing (23 automated tests), repository, README, final report, presentation

Appendix E. How to Run

```
pip install -r requirements.txt
ollama pull qwen2.5:7b-instruct          # once; then open the Ollama app

cd src && uvicorn api:app --port 8000      # grading service, docs at http://localhost:8000/docs

cd web/backend && python manage.py migrate && python manage.py runserver 8001
                                          # teacher application at http://localhost:8001

pytest tests                               # 23 tests, no LLM needed
```

Group 1's digitizer is optional: place their project next to the repository as "Group 1 digitizer" (or set GROUP1_DIGITIZER_PATH) and install Tesseract.